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Role	**Assignee**	**Date**
Approver (Accountable)	Aizik, Elena	10-Jan-2023
Approver (Accountable)	Mustaq, Mohammed	10-Jan-2023
Approver (Accountable)	Afek, Keren	09-Jan-2023
Approver (Accountable)	Kavirayar, Chockalingam	10-Jan-2023
Approver (Accountable)	Habets, Raymond	10-Jan-2023
Approver (User)	Eyal Biran	10-Jan-2023
Approver (Accountable)	Aryev, Anna	09-Jan-2023
Approver (Accountable)	Kliper Goldin, Efrat	10-Jan-2023
Approver (Accountable)	Braunstein, Yonel	09-Jan-2023

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USA Premarket Notification Decision

Advanced Visualization Workspace V15 (ISP13)

1. Introduction 3
2. Premarket Notification Decision 4
2.1. RfD assessment (AVW 15.0.0) 4
3. DEVICE MODIFICATION 5
3.1. Identification of the modified medical device 5
3.2. Description of the AVW15 (ISP13) RfD modifications 6
3.2.1. System Features 6
3.2.2. Rebranding AVW 15 6
3.2.3. CT ASPECTS 7
3.2.4. CT Liver Analysis 7
3.2.5. CT Calcium Automated Analysis 8
3.2.6. CT Comprehensive Cardiac Analysis (CCA) Application 8
3.2.7. Cardiac MR (CMR) Application 9
3.2.8. MR Longitudinal Brain Analysis Application (previously named LoBi) 10
3.2.9. CT Spectral Magic Glass on PACS 11
3.2.10. Photo-Realistic Volume Rendering 11
3.2.11. Multi Modality AVA (MM AVA) 12
3.2.12. Zero Footprint Viewer (ZFP) 12
3.2.13. Brain Perfusion application (BP) 13
***.3.2.14**Discrete Result 13*
3.2.15. NM Suite 13
3.3. Risk-Based Assessment & Design Control Information 14
3.3.1. Risk Based Assessment table 14
3.3.2. Specific Sections 14
4. Conclusion 15
5. References 16
6. Definitions and Acronyms 17
7. Record history 18

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8. Record Review and Approval....18
9. Appendix A: Flowcharts and Decision-making Process....19
9.1. Advanced Visualization Workspace V15 (K162025)....20
9.1.1. Main Flowchart & Questionnaire 20
9.1.2. Labelling Change Chart & Questionnaire 22
9.1.3. Technology, Engineering and Performance Changes Chart & Questionnaire....24
9.1.4. Software Changes Chart & Questionnaire 27
9.1.5. Conclusion 30

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1. Introduction

Philips Medical Systems Nederland B.V. ("Philips" or "the Company") proposes to make modifications to the company's 510(k)-cleared IntelliSpace Portal Platform (ISPP) software medical device (K162025) and to some of its cleared related applications (see section 3.1), as documented below in this USA Premarket Notification Decision memorandum to file. Other than the changes listed below in **Section** 3 of this document, and changes previously documented in the company DHF repository, no other changes have been made to the ISP software device and its applications.

The purpose of this memorandum is to assess these changes introduced as part of Advanced Visualization Workspace V15 (previously named IntelliSpace Portal V13), and document the effect of the changes in accordance with 21 CFR § 807.81 and FDA's two guidance documents issued on October 25, 2017 "*Deciding When to Submit a 510(k) for a Change to an Existing Device*"[1] ("FDA Device Modification Guidance") and *"Deciding When to Submit a 510(k) for a Software Change to an Existing Device*"[2] ("FDA Software Modification Guidance"), and as per the procedure detailed in Philip's US Premarket Submission Procedure ICAP-P-03421.

Per FDA's regulations, manufacturers need to submit a new 510(k) only when a change, or the sum of the incremental changes, exceeds the §807.81(a)(3) threshold, "could significantly affect the safety or effectiveness of the device." Based on FDA's guidance documents and Philip's US Premarket Submission Procedure ICAP-P-03421, Philips obtained the Premarket Notification Decision concluded below in **Section** Error! Reference source not found..

This document provides the decision-making process and final decision made for both types of changes, i.e. significant and non-significant changes. A significant change will necessitate a new 510(k) submission, while a non-significant change will require Internal documentation according to ICAP-T-034021.01/ ICAP-T-072002.07, as applicable. For changes that require Internal documentation only, Philips determined that these proposed modifications could not significantly affect the safety or effectiveness of the device or constitute major modifications under the boundaries cleared originally by FDA, as discussed further below. Therefore, the proposed modifications that were deemed as non-significant and requires internal documentation, do not require a new 510(k), and these changes are documented in this memorandum to file.

The assessment of the changes introduced in AVW15 (ISP 13) comparing to its predecessor is applied in two phases, according to the product release stages (as per ICAP Configuration Management procedure - ICAP-P-036001): in the RfD (Release for Delivery) stage, the company integrated more prominent changes and their regulatory and clinical impact is thoroughly evaluated below; in the next phase, denoted RC (Release Closed), the company will finalize the AVW15 (ISP 13) release with applying refinements and

[1] Available at:
http://www.fda.gov/downloads/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/UCM5147 71.pdf

[2] Available at:
http://www.fda.gov/downloads/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/UCM5147 37.pdf

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corrections, mainly following implementation and use feedback of the modifications integrated during the RfD stage.

2. Premarket Notification Decision

2.1. RfD assessment (AVW 15.0.0)

Conclusion: As summarized in the rationale below and the information documented in **Section 3** of this document, it was concluded that:

- 510(k) submissions are required for: The CT Liver Analysis application previously cleared under K060937, CT ASPECTS, and CT Calcium Automated Analysis applications which are new applications introduced in AVW15.

- Internal documentation is required for the following list of modified applications/features:
  - Product Rebranding
  - Applications:
    - CT Comprehensive Cardiac Analysis (CCA) application
    - MR Cardiac suite (CMR) Application
    - MR Longitudinal Brain Analysis application (previously named MR Longitudinal Brain Imaging (LoBI)
    - MM AVA/sAVA application
    - CT Spectral Magic Glass on PACS application
    - Brain Perfusion application (refer to D000848214 ISP 12.1.6 for the regulatory assessment)
    - Zero footprint viewer (ZFP)
    - NM Suite
  - Features:
    - Discrete Result Sharing
    - Photo-realistic Volume Rendering

Rationale: Modifications introduced to the CT liver analysis application, are an update to the most recent 510(k) as cleared by the U.S. FDA on June 05, 2006 as part of the Brilliance Volume device (K060937). Since the modifications for the CT liver analysis include significant technology updates, as detailed in section 3.9 below, this change requires a 510(k) submission for approving the modifications made. Following this decision CT liver analysis application has been submitted and is under FDA review (K222233). CT ASPECT Scoring and CT Calcium Automated Analysis (CAA) are new applications which will be newly introduced in AVW15, and therefore triggers 510(k) submission.

CT ASPECTS was submitted (K213784) and is under FDA review process, CAA application will be submitted to the FDA in Q12023.

All other modifications, i.e. modifications made to the Advanced Visualization Workspace (IntelliSpace Portal) Platform and modifications made to its related applications, have been summarized and assessed in section 3.2.

Applicable 510(k) Assessment Flowcharts, based on FDA's October 25, 2017 Guidance Documents, "Deciding When to Submit a 510(k) for a Change to an Existing Device" and " Deciding When to Submit a

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510(k) for a Software Change to an Existing Device", were completed (see section 910) comparing the modified ISP and its related applications to the most recently cleared devices.

The following sections provide a risk-based assessment and a detailed evaluation based on the flowchart and text from the guidance documents. Based on the assessment made it was concluded that the modifications to the stand-alone software medical device, i.e. the IntelliSpace Portal Platform (ISPP) and its related applications do not affect the safety and/or effectiveness of the device nor the Intended Use / Indications for Use beyond the established clearances boundaries. Additionally, none of the changes made are the subject of a reportable action. Thus, Philips concluded that internal documentation is required for the Advanced Visualization Workspace platform cleared under K162025 & its related applications (other than CT Liver Analysis, CT ASPECTS and CT Calcium Automated Analysis applications).

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3. DEVICE MODIFICATION

3.1. Identification of the modified medical device

IntelliSpace Portal now rebranded to Advanced Visualization Workspace (AVW), is a software medical device that allows multiple users to remotely access clinical applications from compatible computers on a network. The system allows networking, selection, processing and filming of multimodality DICOM images. This software is for use with off-the-shelf PC computer technology that meets defined minimum specifications.

The Advanced Visualization workspace communicates with imaging systems of different modalities using the DICOM-3 standard.

The Advanced Visualization workspace provides a variety of clinical applications to process clinical images from different modalities (CT, MR, NM, PET, SPECT, Ultrasound, iXR and DXR) that provide advanced visualization capabilities. The applications cover a wide range of clinical domains: oncology, neurology, cardiology, vascular, pulmonary and more.

Advanced Visualization workspace (AVW) including its predecessors have been in the market for over 15 years. Being a Software as a Medical Device (SaMD), the AVW does not have any direct body interaction. AVW's risk management, supported by the PMS data, demonstrates it is a well-established device with a well-known risk profile.

Modified device: Advanced Visualization workspace V15 (IntelliSpace Portal V13)

List of related 510(K) clearances:

- K162025 IntelliSpace Portal Platform
- K052078 Achieva, Intera and Panorama 1.0T Release 2
- K063559 Achieva, Intera and Panorama 1.0T Release 2.5
- K031815 Intera Sensation Release 1 Family
- K060937 Brilliance Volume
- K150665 Spectral CT Applications
- K162955 Multi Modality Tumor Tracking (MMTT)
- K162484 Lung Nodule Assessment and Comparison Option (LNA)
- K042293 Brilliance CT, Private Practice CV configuration
- K182716 CT Brain Perfusion application
- K163250 Longitudinal Brain Imaging (LoBI) application
- K173467 Advanced Diffusion Analysis (ADA) application
- K030520 INTERA FAMILY

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- K130278 MR Permeability
- K203216 The Multimodality Advanced Vessel Analysis (NM AVA)
- K111024 Jetpack 2.0
  K111336 EBW NM 2.0

3.2. Description of the AVW15 (ISP13) RfD modifications

Following is a list of modifications in AVW15 (ISP13) system and applications for evaluation as per D001253823 AVW 15 (ISP13) Technology assessment report

3.2.1. System Features

The following hardware, infrastructure and architecture enhancements are introduced as part of the AVW 15 version:

- System Performance and reliability enhancements
- Visual studio 2019 C# Compiler upgrade.
- Workstation on Windows 10 IoT Embedded LTSC OS (1809 Build)
- Bug fixes related to Security and privacy
- .Net Framework upgrade from .Net 4.7.1 to .Net 4.8
- GPU hardware on AVW server
- Harmony Infrastructure - refactoring of the existing UADE and Graphene infrastructure (note these components are already used for existing LoBi and Liver Health applications). This is a minor modification to enhance the maintainability of the infrastructure.

An impact analysis on the performance and reliability of the product was conducted and documented as per D001253823 AVW 15 (ISP13) Assessment of Technology, Performance and Engineering Changes.

Verification tests have been performed on the new System's design which demonstrated that there is no impact on safety or effectiveness of the device due to these enhancements that were made to the system's infrastructure.

These modifications have no impact on application users, with no impact on the intended use or indications for use and therefore does not require a 510(k) premarket notification. Furthermore, there were no significant changes identified in the following areas for AVW 15, Network communication, Imaging communications standards, Internet standards and VMware / virtualization.

The overall changes introduced to the AVW 15, which was cleared under 510(k) number K162025, were evaluated per the FDA guidance documents flowcharts and decision-making process, as shown in section 9.1 below, concluding that these modifications do not require a 510(k) premarket notification

3.2.2. Rebranding AVW 15

IntelliSpace Portal product name was rebranded to Advance Visualization Workspace due to marketing business decision in order to align with Philips taxonomy. In addition, some minor modifications in applications names were done, mainly for clarification, alignment with catalog and to avoid using applications abbreviations and indicating the full name of the applications.

Updating the product and applications names are considered as a commercial change without any impact on the indications for use, contraindications and clinical functionalities.

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Thus, these updates have no impact on the safety and effectiveness of the device and are not considered a substantial change.

The table below reflects the rebranding the product and application names:

Original full name	**New formal name**
Philips IntelliSpace Portal	Advanced Visualization Workspace
Philips IntelliSpace Portal Server	Advanced Visualization Workspace Server
Philips IntelliSpace IX Workstation	Advanced Visualization Workspace Workstation
Philips IntelliSpace Portal Enterprise	Advanced Visualization Workspace Enterprise
HeartBeat-CS	CT Calcium Scoring
CT Pulmonary Artery Analysis (PAA)	CT Pulmonary Artery Analysis
MR Advanced Diffusion Analysis (ADA)	MR Advanced Diffusion Analysis
MR Longitudinal Brain Imaging (LoBI)	MR Longitudinal Brain Analysis
MR iView BOLD	MR Brain Function Analysis

3.2.3. CT ASPECTS

CT ASPECT Scoring is a SaMD intended to assist the clinician in the assessment and characterization of brain tissue abnormalities using non-contrast CT (NCCT) images according to the Alberta Stroke Program Early CT Score (ASPECTS) for adult (>18 years), acute (≤ 6 hours from onset) stroke patients with a known Large Ventricular Occlusion (LVO) such as a Mid Cerebral Artery (MCA) or Internal Carotid Artery (ICA) occlusion.

The application is assessed as a new application and triggers 510(k) submission which is in process (refer to K213784).

3.2.4. CT Liver Analysis

CT Liver Analysis is a post-processing application providing tools to assess the whole liver, hepatic and portal vasculature, liver segments as well as user-defined regions of interest (ROIs). The application supports the user in the analysis and assessment of quantification parameters of the liver, liver segments as well as user-defined ROIs from liver CT imaging.

Comparing to the CT Liver Analysis in the previously cleared version (K060937), the proposed CT liver analysis application identified the following updates, which are focused on the generic / specific key functionality or workflow.

- Added a new CNN-based liver segmentation algorithm.
- Added a Model-based algorithm for vessel segmentation
- Editing tools for segmentation contour correction

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Following these modifications, as part of the continuous assessment it has been determined that the changes described above in addition to accumulated changes that were made to CT Liver Analysis application since the original submission (ISP platform (K060937) require a 510(K) submission, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in section 9.1.

Accordingly, CT Liver Analysis application has been submitted and is in review process state (K222233).

3.2.5. CT Calcium Automated Analysis

The Advanced Visualization Workspace CT Calcium Automated Analysis application a non-invasive post-processing software intended to evaluate calcified plaques in the coronary arteries, which indicate the presence of coronary artery disease.

The application is assessed as a new functionality and triggers 510(k) submission.

3.2.6. CT Comprehensive Cardiac Analysis (CCA) Application

The application is used to assess the state of the coronary arteries and to analyze functional heart data. The Comprehensive Cardiac Analysis (CCA) application is intended to assist the user in viewing, processing and quantifying cardiac and coronary CTA datasets of patients with suspected or diagnosed cardiac disease including coronary artery disease, for the analysis of heart and coronary artery anatomy and cardiac function. CT Cardiac Plaque Assessment, an option within the CT CCA application, helps physicians determine the presence, extent, and properties of coronary plaques.

As part of AVW improvements the following CCA workflow modifications are introduced in AVW 15:

- Finding Management- a mechanism to capture and manage existing measurements findings for stenosis assessment, including the possibility to export the findings to external report systems. Finding management is a workflow optimization that enables integrated view of coronary findings in a signle view for convenient presentation and communication of results. This change does not introduce any new control mechanisms, nor does this functionality significantly change the use of the device nor does it introduce any new risks. Therefore, no additional clinical data is required. Verification and validation have been demonstrated, there were no unexpected issues and therefore a documentation update is sufficient.
- Multi-batch – a workflow enhancements that enables to save multiple batches at once instead of saving one batch at a time (as in previous ISP version), that is reducing the number of steps in existing a workflow. The selection of batches is predefined by the user. The multi-batch uses the applications batch functionality to allow automatically batch all verified findings to a destination of choice. By design this functionality only sends verified results so there is no new clinical functionality and no additional risk involved.
- CAD-RADS reporting templates- addition of CAD-RADS reporting functionality which enables sharing of clear communication of results in a standardized manner. The user manually controls the presentation of the CAD RADS reporting results. The CADS-RADS functionality provides a workflow to put the existing measurements in a digital copy of the CAD-RADS standard report. Copying existing data electronically into a standard report does not change the risks related to the fact that these measurements could be wrong or misinterpreted as already described in our risk management matrix (D000946860 AVW15 RMM CT Cardiac).

These above updates are aimed at improving the user experience and workflow for coronary CTA analysis and reporting, without significate impact on the clinical safety and efficiency. There is no change in the

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labeling of the application, (according to section 9.1.2.). These design changes are related to integration of exiting platform functionality and a more automatic workflow with better user experience. Thus, these updates are not considered a significant change.

The application has undergone the required V&V activities which were successfully completed. The Comprehensive Cardiac Analysis application has been found to be safe and effective. The software changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device as compared with the cleared K060937 and do not raise any new safety and /or effectiveness issues. These modifications have no impact on application users in addition to the intended use or indications for use and therefore do not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in appendix A section 9.1.

3.2.7. Cardiac MR (CMR) Application

MR Cardiac is a suite of applications in the Advanced Visualization Workspace 15, which contains the MR Cardiac Viewer and the following analysis applications: Functional Long Axis, Functional Left Ventricle & Right Ventricle, Temporal Enhancement, Spatial Enhancement and Quantitative Mapping previously cleared under (K063559).

The MR Cardiac Suite modifications newly introduced in AVW15 mainly focuses at improving the user experience and workflow creating an all-in-one environment for cardiac analysis with an emphasis on increasing user's throughput of results and reports generation. Additionally, the MR Cardiac requirements were rephrased for clarifications with no additional changes to core algorithms and clinical functionality.

The main changes are:

- MR Cardiac Viewer- in addition to MR data, the viewer will also support multi-modality viewing (CT, US, XA, NM, PET, CR, and DX). This is a minor change without new clinical functionality or change to the intended use or indications for use, the multi-modality viewing capabilities were previously cleared under K162025. It does not introduce any new control mechanisms, nor does this functionality significantly change the use of the device nor does it introduce any new risks. Therefore, no additional clinical data is required. Verification and validation have demonstrated there were no unexpected issues and therefor a documentation update is sufficient.
- Workflow and UI enhancements intended for better user experience, with an emphasis on increasing user's throughput of results and reports generation. The enhancements include:
    - The ability to launch external application- to facilitate access and activate external applications directly from MR Cardiac Suite UI for completion of the MR Cardiac reading within same user workflow (e.g. ability to launch the existing MR QFlow application; cleared under K003459, K031815).
    - Finding dashboard is a technical component consolidating the existing measurements of the CMR Suite of applications into one place, enabling structured results export for reporting. The findings dashboard also provides an ability to define customized notification mechanism on reading results per user preferences.
    - New "workflow" feature allows users to define their own reading workflow according to their reading preferences. Providing improved accessibility to CMR applications.
    - Side by side display of CMR applications within MR Cardiac Suite.
    - Prefetch existing prior studies from patient directory for viewing and compression with current studies.
    - User configurable automatic labeling for image type and orientation.

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The above changes aimed at improving the user experience and workflow, without impact on the clinical safety nor significant impact on efficiency. Since the application does not introduce any new control mechanisms, nor does this functionality significantly change the use of the device, nor does it introduce any new risks; no additional clinical data was required. Usability analysis was performed to assess the impact of the changes on usability related risks. Verification and validation have demonstrated, there were no unexpected issues and therefore a documentation update is sufficient.

- Updated editing tools in MR cardiac suite - additional manual tools were added to address the user need for editing contours more conveniently. All the editing tools are fully controlled by the user to be used for manual contour drawing. This implementation change aims at a better user experience. It does not introduce any new control mechanisms, nor does this functionality significantly change the use of the device nor does it introduce any new risks. Therefore, no additional clinical data is required. Verification and validation have demonstrated there were no unexpected issues and therefor a documentation update is sufficient.
- Additional ability to import existing "short axis" contours generated in Functional LV/RV to other short axis applications (Temporal Enhancement and Mapping applications). The change aims to save additional contouring time in these applications. The user is able to manually edit the final output. This change has no impact on the application's clinical functionality and has no significant impact on performance specifications.

The application has undergone the required V&V activities which were successfully completed. The MR Cardiac Suite has been found to be safe and effective. The software changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device as compared with the cleared ACHIEVA, INTERA and PANORAMA 1.OT Release 2.5-series (K063559), and do not raise any new safety and /or effectiveness issues. In summary, these modifications have no impact on applications' the intended use /indications for use and therefore are eligible for internal filing and do not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in appendix A section 9.1

3.2.8. MR Longitudinal Brain Analysis Application (previously named LoBi)

The Longitudinal Brain Analysis application (K163250) is a post-processing application to be used for viewing and evaluating neurological images provided by a magnetic resonance diagnostic device. The application is intended for viewing, manipulation, 3D- visualization and comparison of medical imaging and/or multiple time-points. The application enables visualization of information that would otherwise have to be visually compared disjointedly. The application provides analysis tools to help the user assess, and document changes in diagnostic and follow-up examinations. The application is designed to support the workflow by helping the user to confirm the absence or presence of lesions, including evaluation, quantification, follow-up and documentation of any such lesions. The physician retains the ultimate responsibility for making the final diagnosis and treatment decision.

The update introduced in MR Longitudinal Brain Analysis application is to provide a hanging protocol presentation of FLAIR* image type generated using multiplication of existing Flair and T2* images from the same study. It is an additional viewing option presented next to Flair and T2* images, enhancing existing capabilities, thus no impact on the application's clinical functionality, its performance specifications, and the indications for use. This functionality does not significantly change the use of the device, nor does it introduce any new risks, therefore, no additional clinical data is required.

The application has undergone the required V&V activities which were successfully completed with no unexpected issues. The Longitudinal Brain Analysis has been found to be safe and effective. The software

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changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device as compared with the cleared Longitudinal Brain Analysis K163250 and do not raise any new safety and /or effectiveness issues. This modification has no impact on the intended use or indications for use and therefore does not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in Appendix A section 9.1.

3.2.9. CT Spectral Magic Glass on PACS

The CT Spectral Magic Glass on PACS (sMGoP) provides a set of tools to review and analyze Philips spectral CT datasets directly within the PACS viewer (K150665).

The main change introduced in the CT Spectral Magic Glass on PACS application is enabling the 3D viewing functionality of the AVW platform such as MPR and 3D volume rendering as in other CT review applications (K162025). This change does not introduce any new control mechanisms, nor does this functionality significantly change the use of the device nor does it introduce any new risks. Therefor no additional clinical data is required. Verification and validation have demonstrated there were no unexpected issues and therefor a documentation update is sufficient.

The application has undergone the required V&V activities which were successfully completed. The Longitudinal Brain Analysis has been found to be safe and effective. The software changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device as compared with the cleared spectral magic glass on PACS K150665 and do not raise any new safety and /or effectiveness issues. This modification has no impact on the intended use or indications for use and therefore does not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in appendix A section 9.1

3.2.10. Photo-Realistic Volume Rendering

The Photo Realistic Volume Rendering (PRVR) engine is a photo-realistic post-processing technique used in 3D visualization of medical images. It provides best image quality of high density and high-contrast structures such as bones and contrast enhanced vessels. The Photo Realistic Volume Rendering (PRVR) has an artificial light source that delivers lifelike 3D images and gives the operator the ability to move the light source anywhere in the 3D volume.

The Photo Realistic Volume Rendering (PRVR) photo-realistic 3D volume rendering mode can be used on volumetric viewports in both the CT and Cardiac Viewers. The spectral viewers were now extended to include the platforms photorealistic 3D rendering with no effect on clinical functionality or performance specification. This mode has the same intended use as the other volume rendering techniques that already exist in the previous ISP12 version and does not require additional data as this functionality is not deemed to be for diagnostic information.

All volume rendering viewers including the new Photo Realistic Volume Rendering (PRVR) are non-diagnostic.

The application has undergone the required V&V activities which were successfully completed. The PRVR application has been found to be safe and effective. The "photorealistic" appearance is clinically equivalent to the existing volume rendering solutions as compared with the cleared ISPP (K162025) and therefore

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does not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. This modification has no impact on application users in addition to the intended use or indications for use and therefore does not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in appendix A section 9.1

3.2.11. Multi Modality AVA (MM AVA)

The Multi-Modality Advanced Vessels Analysis (MM AVA) application is intended to assist the user in viewing, processing and quantifying vascular conventional or spectral CTA and MRA datasets of patients with suspected or diagnosed vascular diseases, for the analysis of vessel anatomy. It provides a unified workflow for CT Angiography (CTA), MR Angiography (MRA) modalities and now as well for spectral data. For CTA data, it provides both automatic and manual bone removal and vessels segmentation including extraction of vessel centerlines, lumen contours and vessel contours. For all modalities, it provides tools for extracting and editing centerlines.

MM AVA offers inspection views for selected vessels centerlines and local analysis. It allows creating, capturing, and reviewing of basic user selected endovascular measurements (and calculations when applicable), as well as predefined measurements sets and measurements correlations.

The MM AVA application was the cleared under (K203216). The update introduced under AVW15 is to enables spectral capabilities (cleared under K150665) and to read spectral data through MM AVA without any changes to the clinical functionality nor clinical algorithms. These updates are not considered a significant change and does not introduce new functionality no further clinical data and do not impact on the safety and /or effectiveness of the product. Indication for use of sAVA and AVA were combined with respect to supported datatypes.

The application has undergone the required V&V activities which were successfully completed. The Multi Modality AVA (MM AVA) has been found to be safe and effective. The software changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device as compared with the cleared MM AVA (K203216 and K150665) and do not raise any new safety and /or effectiveness issues. This modification does not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in appendix A section 9.1

3.2.12. Zero Footprint Viewer (ZFP)

The Zero Footprint Viewer (ZFP) is a web-based viewer (thin client) intended to be used by trained professionals, including but not limited to radiologists. The Zero Footprint extends the availability of images that are available on IntelliSpace Portal to peer radiologists, other non-radiology physicians or referring physicians. Zero Footprint Viewer is not used for diagnosis. The software offers digital image processing, measurements, manipulation and quantification of images and communication. Images are accessible to any web browser on a computer. In AVW15, ZFP uses the updated viewing infrastructure as introduced with the MR Cardiac Suite.

The application has undergone the required V&V activities which were successfully completed. The Application has been found to be safe and effective. The software changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device as compared with the cleared K162025 and do not raise any new safety and /or effectiveness issues. This modification has no impact on the intended use or indications for use and therefore does not require a 510(k) premarket

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notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in appendix A section 9.1

3.2.13. Brain Perfusion application (BP)

BP application is a post processing software application intended to assist with the evaluation of an area of interest, to generate qualitative and quantitative information about changes in image intensity over time. It supports the analysis of dynamic/serial CT after injection of contrast agent, by calculating the parameters related to brain perfusion and displays the results as a composite (single image that is calculated from a set of time course images at a single location) image.

Updates were introduced as part of ISP12.1.6, it was assessed under ISP12 Regulatory Assessment (D000848214) document.

3.2.14. Discrete Result

Discrete results is a new workflow enhancement for efficient AVW result sharing (reporting) with 3rd party health systems, thus, increasing efficiency in reporting and communication between users. This feature is available for the following applications: CCA, TAVI, Calcium scoring, Cardiac MR and LNA.

These changes enhance the user experience, but are not significant in terms of clinical functionality. The application has underwent the required V&V activities which were successfully completed. The Application has been found to be safe and effective. The software changes do not significantly affect clinical functionality or performance specifications that are directly associated with the intended use of the device. These changes do not significantly affect the safety or effectiveness of the device and do not raise any new safety and /or effectiveness issues. This modification has no impact on the intended use or indications for use and therefore does not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in section appendix A section 9.1

3.2.15. NM Suite

NM Suite is a SaMD (software as a medical device) which is launched using the AVW15 environment. The NM Suite includes NM Jetpack and NM Planar applications (covered by K111024 and K111336), which were previously part of ISP device. There were no modifications or changes in their functionality.

NM Suite implements State of the Art clinical methods and provide for physicians a viewing, processing and analysis environment for Planar data and usage of specific Spect Nuclear Medicine data as well.

NM Planar and Jetpack applications, are divided by anatomical regions: cardiac, Lung, endocrine/glands, renal, gastrointestinal/esophagus, hepatobiliary/liver, whole body/bone/MIBG/lymph nodes and brain. The NM Suite offers display utilities and non-diagnostic quality assurance utilities that assist with the calibration of the nuclear medicine imaging systems.

NM Suite is launched from AVW environment and is part of its installation package (deployed in a single iso file). Positioning of NM Suite as a separate medical device was done due to a business decision.

The separation from AVW is reflected in the documentation and device label only and has no impact on the product design, requirements, deployment nor user interface. Except for the device label this change is transparent to the users. As no design changes were made since the previous 510K submission which NM Suite was covered by (K111024 2.0 Jetpack, K111336 EBW NM 2.0), these updates are not

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considered a significant change and does not introduce any new functionality, no further clinical data and do not impact on the safety and /or effectiveness of the product.

The application has undergone the required V&V activities (as part of AVW15 release) which were successfully completed. The Application has been found to be safe and effective. No new safety and /or effectiveness issues were raised. This modification has no impact on the indications for use and therefore does not require a 510(k) premarket notification, as was concluded per the FDA guidance documents' Flowcharts and decision-making process documented in section appendix A section 9.1

3.3. Risk-Based Assessment & Design Control Information

A risk-based assessment has been conducted to determine the impact of the changes on AVW15 safety and effectiveness according to FDA guidance documents “Deciding when to Submit a 510(k) for a change to an existing device” and “Deciding when to submit a 510(k) for a software change to an existing device (October 2017)”. The assessment involved analyzing possible sequences of events, hazardous situations, and associated possible harm with regard to performed modifications. The extent of risks and harms associated with modifications were assessed, individually and in aggregate. The assessment also involved the evaluation whether the modifications degrades the device effectiveness. The assessment is documented in the spreadsheet below.

3.3.1. Risk Based Assessment table

Worksheet AVW15
USA premarket notif

3.3.2. Specific Sections

User facing labeling updates

AVW15 IFU kit content (300008780681- RfD) was updated to include instructions for use for the enhanced features, changes, included in AVW15 release. These are within the scope of the intended use and indications for use of the device and do not significantly affect the safety or effectiveness of the device. In addition, the IFU kit content also includes CT ASPECTS and NM Suite IFUs.

New or Modified Architecture and Infrastructure

AVW15 software architecture was developed in a planned, modular format. Changes introduced to AVW15 were according to the DHF380234 System Design Specifications and were done in a limited and modular format, reducing the likelihood of unintended consequences/impact to other areas of the code. AVW15 has been successfully verified following the modifications (D000996308 Verification Report). No complex infrastructure change to the device. There are no changes to the overall structure of the software. These changes do not significantly affect the safety or effectiveness of the device.

New or Modified “Reengineering” and “refactoring”

Refactoring modifications to enhance the maintainability of the device within its specifications have been conducted on select areas. Those changes are of narrow scope and limited in complexity. The code has

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been reviewed and assessed as part of the development of the product. AVW 15 has been successfully verified following these modifications (D000996308 Verification Report). No significant software re-write. The device has not undergone re-engineering. These changes do not significantly affect the safety or effectiveness of the device.

4. Conclusion

The modifications of the AVW15 (ISP13) and its related applications have been assessed in the technology assessment document (D001253823) and in Appendix A of this document as per 510(k) Assessment Flowcharts, based on FDA's Guidance Documents ("Deciding When to Submit a 510(k) for a Change to an Existing Device" and " Deciding When to Submit a 510(k) for a Software Change to an Existing Device". The effects of labeling, SW, technology, engineering, and performance changes were evaluated:

- Since the modifications for the CT Liver Analysis included significant commutative changes, as detailed in section 3.2.2 and 9.1, it was determined as requiring a 510(k) submission for clearing the modifications. In addition, CT ASPECTS and CT Calcium Automated Analysis since the applications are new and triggers a 510(k).
- Modifications for the AVW15 (ISP13) and its related application (except CT Liver, CT ASPECTS and CT Calcium Automated analysis) are enhancements, addition of features to existing applications to address the customers' needs and field feedback as identified in the URS (DHF380082), they do not include significant updates as detailed in sections 3.2.1, 3.2.3-3.3.15. The changes are not done due to a recall or corrective action. The analysis has shown that the changes do not significantly modify or impact the overall safety and effectiveness of the cleared device. None of these changes affect the indications for use. No additional clinical data is required to evaluate Safety and Efficacy and the validation results do not raise issues of Safety and Efficacy. The intended use of the AVW15 platform remains unchanged. Therefore, the modifications to the AVW15, can be based on internal filing.
- Furthermore, Cumulative analysis has shown that the changes do not significantly modify or impact the overall safety and effectiveness of the AVW15 device.
- The overall risk-based assessment details in Section 3.3 demonstrates that the changes do not significantly affect the safety and effectiveness of the AVW15 (ISP13) device.
- Note that none of these changes affect the indications for use (A1), no additional clinical data is required to evaluate S&E (B5.3) and the validation results do not raise issues of S&E (B5.4), see Appendix A.

Therefore, based on this evaluation and the reference(s) listed in section 5 of this document, the AVW 15 (ISP13) has no change in the indications for use and the changes do not significantly affect the safety and the effectiveness. Therefore, a premarket notification submission to the FDA is not required for these modifications, aside from the changes introduced for CT Liver Analysis. In addition, for CT ASPECTS and CT Calcium Automated analysis since the applications are new applications.

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5. References

Reference	Document Title	Document ID Number	Revision Number
Ref -1	URS Requirements Specification of [Product name]	DHF380082	LF
Ref -2	510k clearance of Product Name	N/A	N/A
Ref -3	Technology Change Assessment of AVW15 (ISP13)	D001253823	A
Ref -4	AVW15 (ISP13) Design Verification Report	D000996308	D
Ref -5	AVW15 (ISP13) PRS	DHF380211	LG
Ref -6	AVW15 (ISP13) SDS	DHF380234	LD
Ref -7	AVW 15 (ISP13) Technology, Performance and Engineering Changes Assessment	D001253823	A
Ref -8	AVW 15 (ISP13) Validation Report	D000926820	C
Ref -9	AVW 15 (ISP13) Risk Management Report	D001207821	A
Ref -10	ISP13 System RMA	DHF380158	LF
Ref-11	ISP13 System RMM	D000996286	C
Ref – 12	ISP NM Suite Application RMA	D001024262	A
Ref -13	AVW 15 RMA CT Cardiac	D000946872	C
Ref -14	ISP 13 RMM CT Cardiac	D000946860	B
Ref -15	ISP13 Legacy MR Cardiac RMA	D001065179	A
Ref -16	ISP13 RMA MRCardiac Suite	D000991970	D
Ref -17	ISP13_RMM_MRAPPS & MMV	D001065108	A
Ref -18	ISP13_Risk Management Analysis_MRApps & MMV	D001065109	A
Ref -19	ISP13 Legacy MR Cardiac RMA	D001065179	A

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Ref -20	ISP13 RMA MRCardiac Suite	D000991970	D
Ref -21	ISP13_RMM_MRAPPS & MMV	D001065108	A
Ref -22	ISP13_Risk Management Analysis_MRApps & MMV	D001065109	A
Ref -23	ISP13 RMA Spectral CT Apps	D000952406	D
Ref -24	ISP13 RMA for CTapps	D000938106	F
Ref-25	ISP13 RMM MRCardiac Suite	D000879224	E
Ref- 26	Risk Management Matrix for ISP13,MR Applications and MMV	D001104871	B
Ref -27	ISP13 RMM SpectralCT Apps	D000996304	D
Ref -28	ISP13 RMM General CTApps	D000892550	F
Ref -39	ISP13 RMA MRCardiac Suite	D000991970	D
Ref -30	RMM for CT Aspects	D001157471	B
Ref -31	CT ASPECTS RMA	D001157472	A
Ref -32	ISP NM Suite Application RMA	D001024262	A
Ref -33	ISP NM Suite Application RMM	D001024288	A
Ref -34	ISP13 RMA LoBI	D001105931	B
Ref -35	CT ASPECTS Design Verification Report	D001157473	A
Ref -36	NM Suite Design Verification Report	D001246601	A

6. Definitions and Acronyms

Definition	Description
N/A	Not applicable

Acronym	Description
N/A	Not applicable
S&E	Safety & Effectiveness

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7. Record history

Rev	Release Date	Author	Description of changes
A	In Windchill	Shir Peleg	Initial revision for AVW15(ISP 13) RFD release

8. Record Review and Approval

Name Author	Function Author
Shir Peleg	Regulatory Affairs Specialist

Name Reviewer(s)	***Function Reviewer(s)***	***Review Date(s)*** *DD-MMM-YYYY*	***Signature(s)***
Keren Afek	Sr. Regulatory Affairs Specialist	In windchill	In windchill
Elena Aizik	Sr. Regulatory Affairs Specialist	In windchill	In windchill
Yonel Braunstein	Regulatory Affairs Head of Regulatory Affairs	In windchill	In windchill
Mustaq Mohammed	Project Manager	In windchill	In windchill
Efrat Kliper Goldin	Product Manager	In windchill	In windchill
Raymond Habets	Clinical Specialist	In windchill	In windchill
Chocks Kavirayar	System Architect	In windchill	In windchill
Anna Aryev	Process QA/QE	In windchill	In windchill

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Name Approver	Function Approver	Date DD-MMM-YYYY	Signature
Eyal Biran	Program Manager	In windchill	In windchill

End of Document

9. Appendix A: Flowcharts and Decision-making Process

The Questionnaire and Flowchart below are according to the guidance "Deciding when to submit a 510(k) for a Change to an Existing Device" (October 25, 2017) and "Deciding when to submit a 510(k) for a Software Change to an Existing Device" (October 25, 2017). Follow the Flowchart to determine if a "yes" in the table mandates a new 510(k) submission. Provide an explanation of the nature of the modification for each response in the table/flowchart below.

Note: Since the IntelliSpace Portal is a software only medical device the changes will be assessed using the main Flowchart, Flowchart A (if labeling changes are applicable) and Flowchart B Technology, Engineering, and Performance Changes (if applicable) of the “Deciding when to submit a 510(k) for a Change to an Existing Device” (October 25, 2017) guidance and "Deciding when to submit a 510(k) for a Software Change to an Existing Device" (October 25, 2017) for software changes assessment.

Flowchart C Material Changes and Flowchart D for IVD Changes, of the “Deciding when to submit a 510(k) for a Change to an Existing Device” (October 25, 2017), are not applicable for the IntelliSpace Portal as it is a Software as Medical Device (SaMD) loaded on off-the-shelf hardware, i.e. it does not contains material, and also is not an IVD device.

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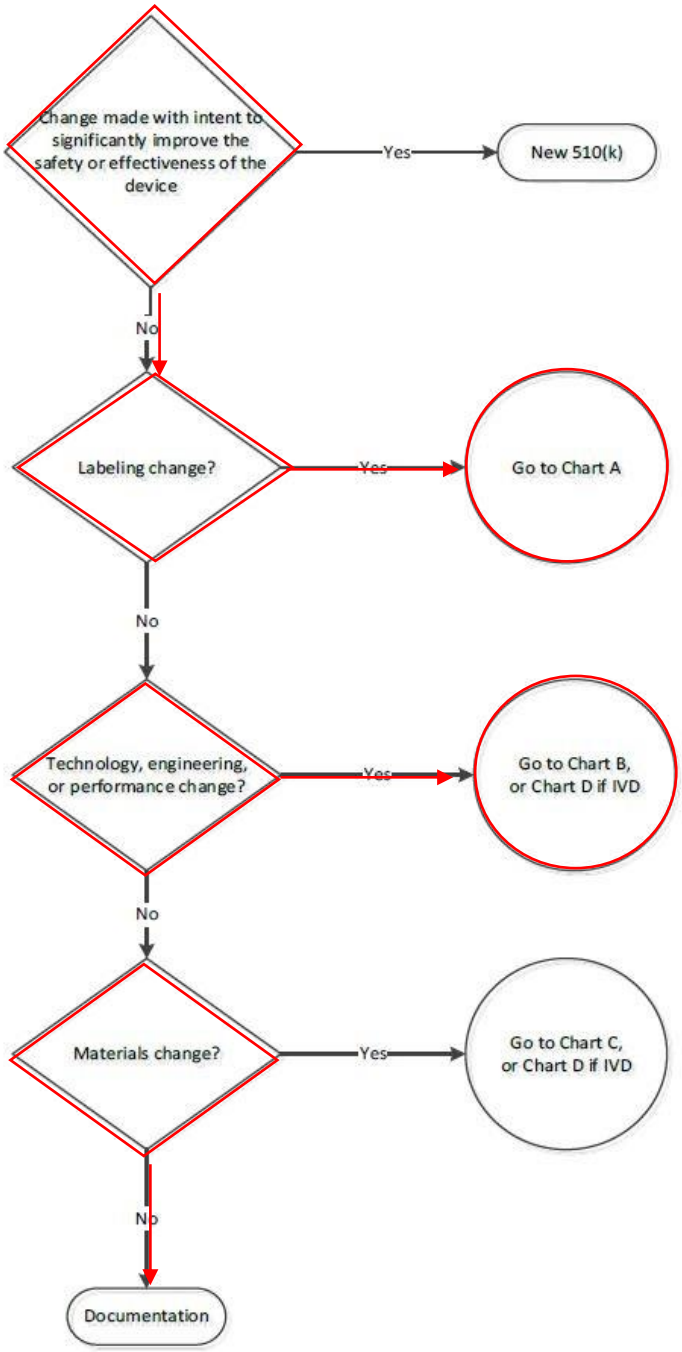
9.1. Advanced Visualization Workspace V15 (K162025)

9.1.1. Main Flowchart[3] & Questionnaire



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#	Question	Yes/No	Decision / Rationale
1	Change made with intent to significantly improve the safety or effectiveness of the device	☒ Yes	**Yes**, for CT Liver Analysis, (introduction of new functionalities and changes since the originally cleared device).
		☒ No	**No**, The modifications in these applications (detailed in section 3.2.6 until 3.2.15) are not intended to significantly affect the safety or effectiveness of the device, but to improve the user experience and customer satisfaction in the usage of the device. Changes are also intended to maintain current with standard Off-the-Shelf hardware and software on which the SaMD is loaded. Refer to section 3.1.2 “System Feature” for risk-based assessment of the individual changes. In conclusion, the modifications implemented do not significantly affect the safety or effectiveness of the device.
2	Labeling Change	☒ Yes ☐ No	If "Yes", Go to Chart A in Section 9.1.2. If "No, section 9.1.2 9.1.2 is N/A
3	Technology, engineering, or performance change?	☒ Yes ☐ No	If "Yes", Go to Chart B in Section 9.1.3 If "No, section 9.1.3 is N/A
4	Materials change?	☐ Yes ☒ No	N/A, as this device is a stand-alone software (SaMD).

[3] Flowchart from FDA Guidance “Deciding when to submit a 510(k) for a Change to an Existing Device” (October 25, 2017).

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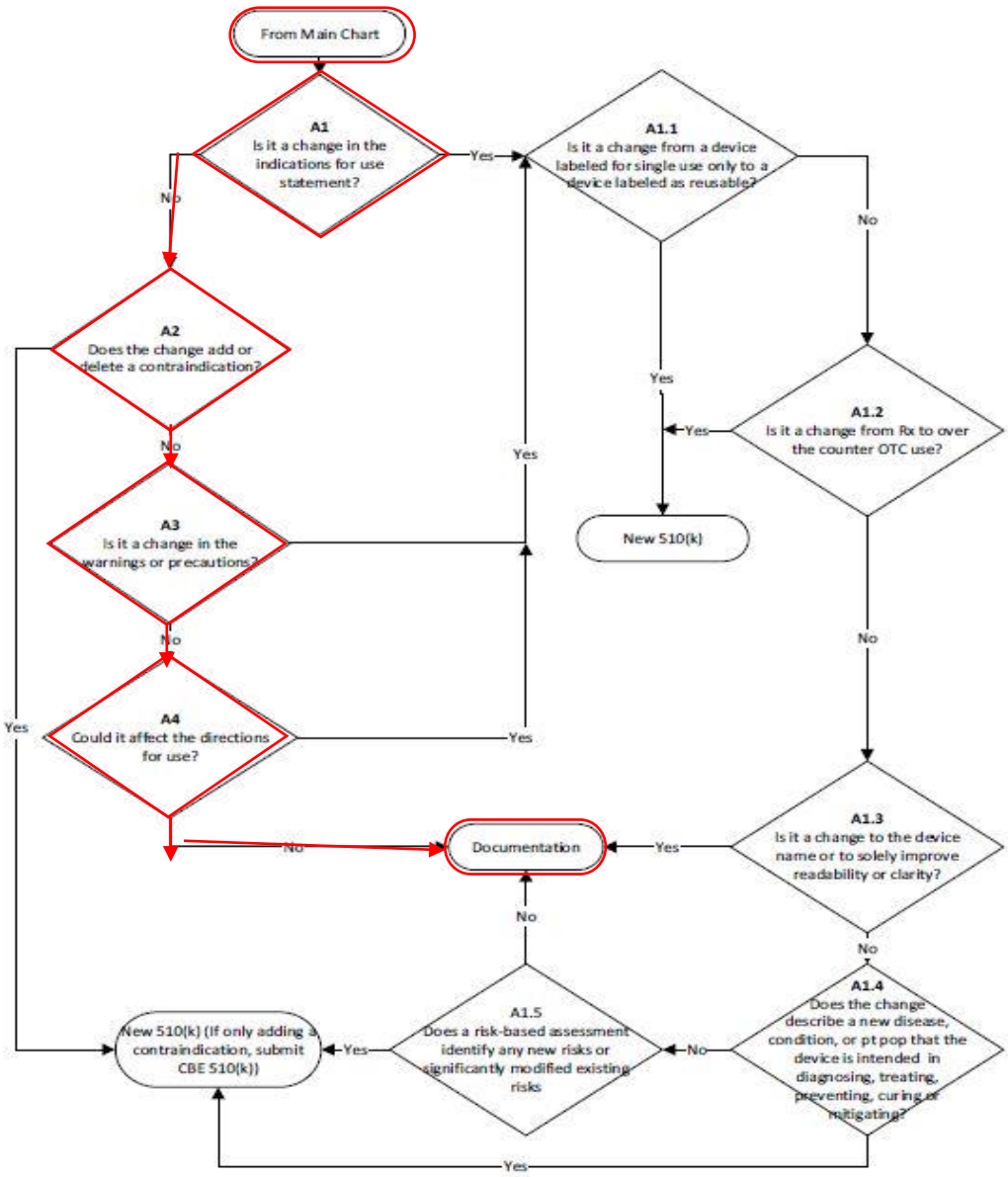
9.1.2. Labelling Change Chart & Questionnaire

Chart A for Labelling Changes[4]



Flowchart A: Labeling Changes

[4] Flowchart from FDA Guidance "Deciding when to submit a 510(k) for a Change to an Existing Device" (October 25, 2017).

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#	Question	Yes/No	Decision / Rationale
A1	Is it a change in the indications for use statement?	☐ Yes ☒ No	No, go to A2 Rationale: There is no change in the Indications for Use of the device
A2	Does the change add or delete a contraindication?	☐ Yes ☒ No	No, go to A3 Rationale: The changes did not require the addition or deletion of a contraindication.
A3	Is it a change in the warnings or precautions?	☐ Yes ☒ No	No, go to A4 Rationale: The changes did not require a change in the warnings or precautions (minor editorial clarifications were implemented).
A4	Could it affect the directions for use?	☐ Yes ☒ No	Yes, go to A.1.1 Rationale: modifications implemented in the labeling do not affect how the device is used in practice. There is no change in the principles of operations and no substantial change in the direction for use.

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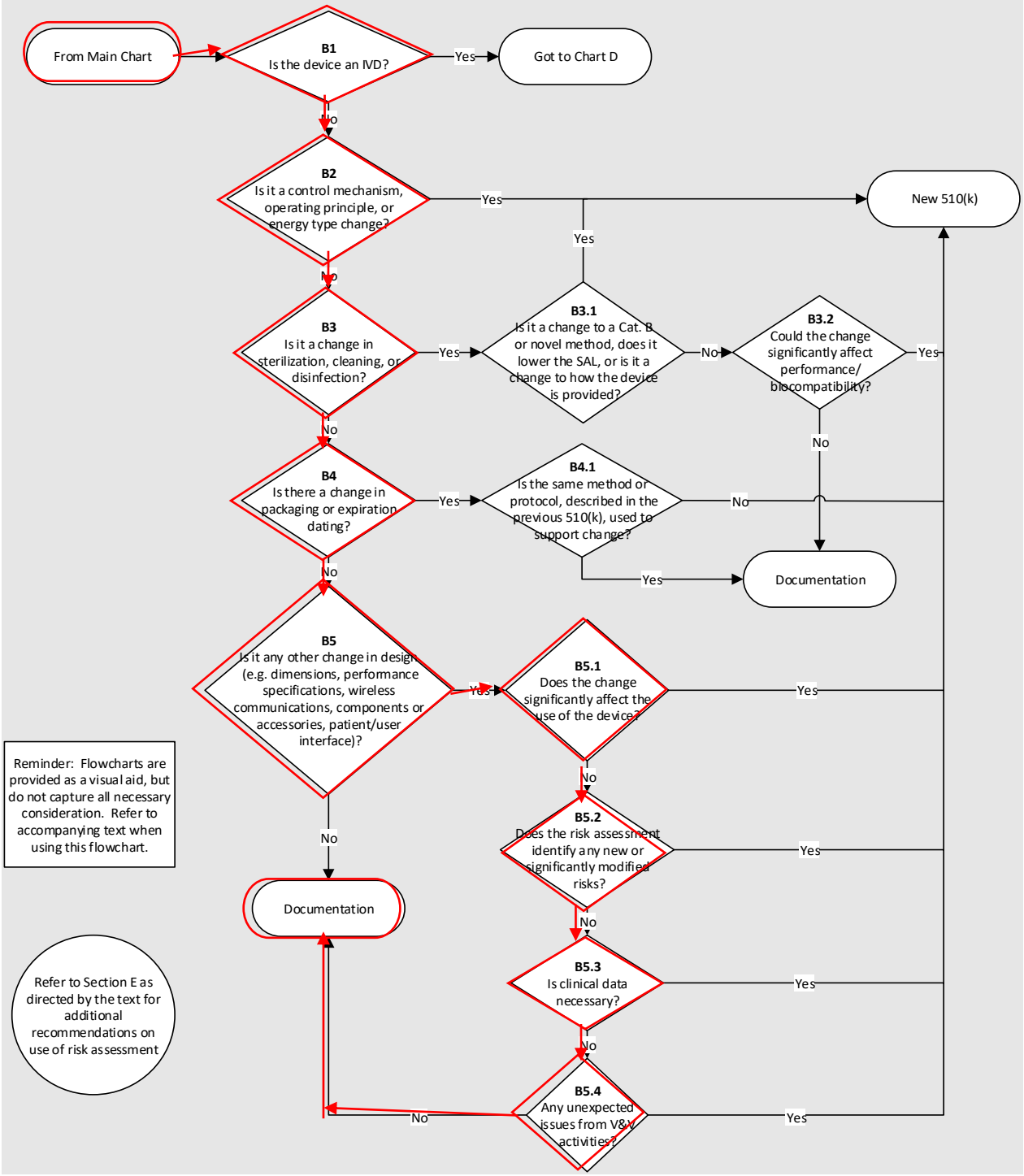
9.1.3. Technology, Engineering and Performance Changes Chart[5] & Questionnaire

Deciding When to Submit a 510(k) for a Change to an Existing Device, issued October 25, 2017
Figure 3 - Flowchart B: Technology, Engineering, and Performance Changes
(Contains Nonbinding Recommendations)



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#	Question	Yes/No	Decision / Rationale
B1	Is the device an in vitro diagnostic device?	☐ Yes ☒ No	The device is a SaMD device and is not an IVD device.
B2	Is it a control mechanism, operating principle, or energy type change?	☐ Yes ☒ No	No, go to B3 Rationale: No, the SaMD modifications do not impact the control mechanism, operating principles, or change the energy type.
B3	Is it a change in sterilization, cleaning, or disinfection?	☐ Yes ☒ No	No, go to B4 Rationale: This is an SaMD device. Thus, sterilization, cleaning, or disinfection are not applicable for the device.
B4	Is there a change in packaging or expiration dating?	☐ Yes ☒ No	No, go to B5 Rationale: No physical Packaging; this is an SaMD device that is downloaded electronically from a website and is not delivered on a physical medium.
B5	Is it any other change in design (e.g., dimensions, performance specifications, wireless communication, components or accessories, or the patient/user interface)?	☒ Yes ☐ No	Yes, go to B5.1 Rationale: Variable software design modifications were implemented in the applications listed in clause 3.2. A detailed description is provided in the same section.
B5.1	Does the change significantly affect the use of the device?	☒ Yes (for CT Liver Analysis) ☒ No (for changes 3.2.6-3.2.15)	No, go to 5.2 Yes, 510(k) required Rationale: As detailed is section 3.2.4, the changes made in the following application significantly affect the use of the device: *CT liver analysis* The changes made in all other applications do not significantly affect the use of the device.
B5.2	Does a risk-based assessment of the changed device identify any new risks or significantly modified existing risks?	☐ Yes ☒ No	No, go to 5.3 Rationale: No, the " Risk-Based Assessment" Table presented in section 3.3 did not identify any new risks or significantly modified existing risks.

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#	Question	Yes/No	Decision / Rationale
B5.3	Are clinical data necessary to evaluate safety or effectiveness for purposes of design validation?	☐ Yes ☒ No	No, go to 5.4 Rationale: No, only non-clinical design verification and validation activities were performed to confirm the device meets it specifications
B5.4	Do design verification and/or validation activities produce any unexpected issues of safety or effectiveness?	☐ Yes ☒ No	No, Documentation Rationale: design verification and validation activities demonstrated that the design meets its pre-defined specifications and did not raise any unexpected issues of safety or effectiveness.

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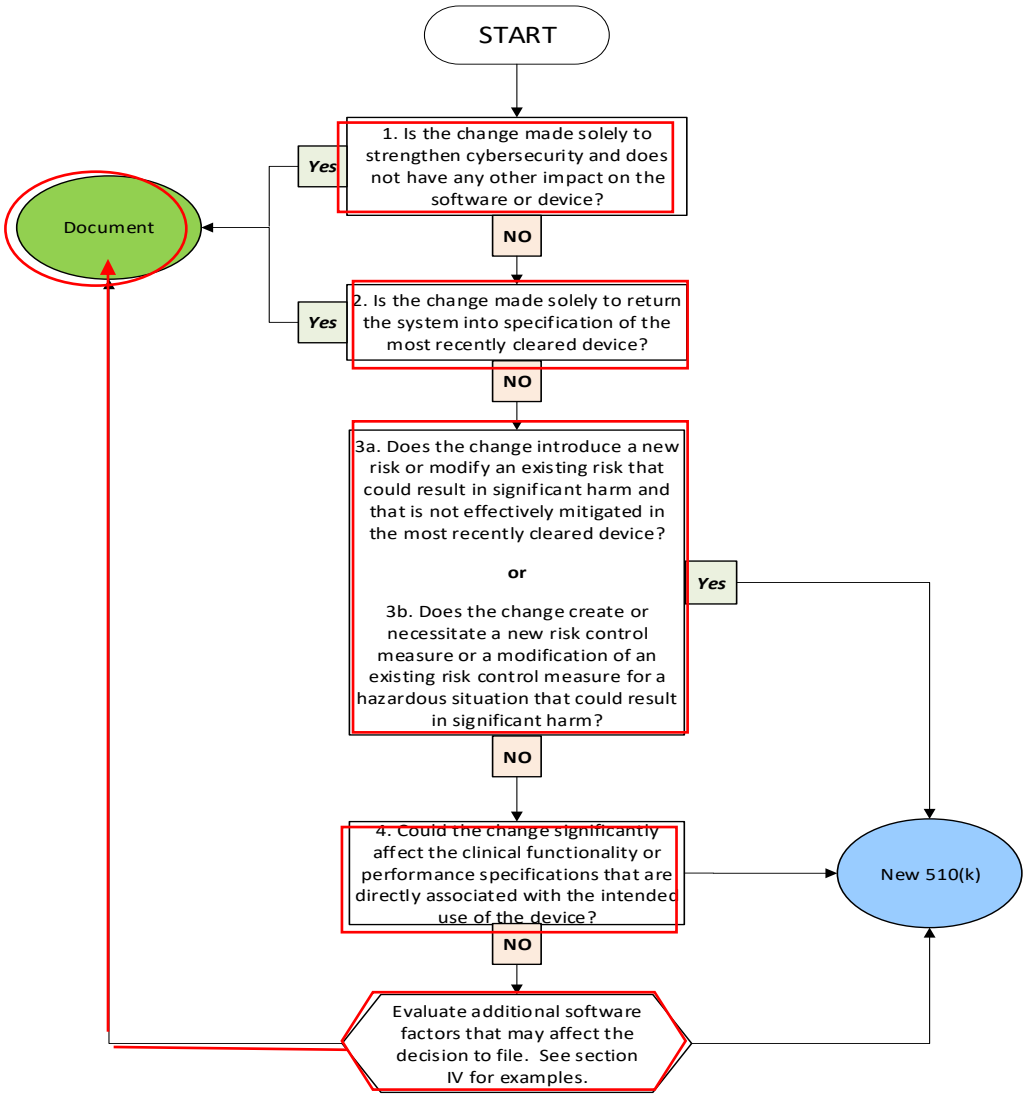
9.1.4. Software Changes Chart6 & Questionnaire

When to Submit for a SW Change

Deciding When to Submit a 510(k) for a Software Change to an Existing Device, issued October 25, 2017
Figure 1 - When to Submit a New 510(k) For a Software Change to an Existing Device.
Contains Nonbinding Recommendations



This flowchart is not indented to be used as a 'stand-alone' document and should only be considered in conjunction with the accompanying text in the guidance.

[6] Flowchart from FDA Guidance "Deciding when to submit a 510(k) for a Software Change to an Existing Device" (October 25, 2017).

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#	Question	Yes/No	Decision / Rationale
1	1. Is the change made solely to strengthen cybersecurity and does not have any other impact on the software or device?	☐ Yes ☒ No	No, go to Question 2 Rationale: No, the change was not made solely to strengthen cybersecurity
2	2. Is the change made solely to return the system into specification of the most recently cleared device?	☐ Yes ☒ No	No, go to Question 3a&b Rationale: No, the change was not made to return the system into specifications of the most recently cleared device
3a	3a. Does the change introduce a new risk or modify an existing risk that could result in significant harm and that is not effectively mitigated in the most recently cleared device?	☐ Yes ☒ No	No, go to Question 3b Rationale: No, as shown in the " Risk-Based Assessment" Table presented in section 3.3, the change did not identify any new risks or significantly modified existing risks.
3b	3b. Does the change create or necessitate a new risk control measure or a modification of an existing risk control measure for a hazardous situation that could result in significant harm?	☐ Yes ☒ No	No, go to Question 4 Rationale: No, the changes enhanced existing risk control measures, while the risk control measures in the most recently cleared devices effectively mitigated the hazardous situation.
4	4. Could the change significantly affect the clinical functionality or performance specifications that are directly associated with the intended use of the device?	☒ Yes (for CT liver analysis) ☒ No, for changes 3.2.6-3.2.15)	No, go to Question 5 Rationale: Yes, changes in CT Liver Analysis application significantly affect the performance specifications which are directly associated with the intended use of the device. No, all other changes optimized the workflow and enhanced system performance, without affecting clinical functionality or performance specifications that are directly associated with the intended use of the device.

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#	Question	Yes/No	Decision / Rationale
5	Evaluate additional software factors that may affect the decision to file.	See next steps	Common Software Change types should be evaluated see listing below. Indicate with Yes/No whether any of these types of modifications are present and provide Trace ID
a	Have any **"Infrastructure"** changes been made (modifications made to the software support system)?	☐ Yes ☐ No	Rationale: No complex changes, refer section 3.3.2
b	Have any **"Architecture"** changes been made (modifications to the overall structure of the software)?	☒ Yes ☐ No	Rationale: refer section 3.3.2
c	Have any **"Core algorithm"** changes been made (modifications made to an algorithm that directly impact or contribute to the device's intended use)?	☒ Yes (for CT liver analysis) ☒ No	Rationale: No, in CCA, CMR, Lobi, AVA, ZFP, NM Suite, applications the change in the algorithms does not constitute a change in basic functionalities. Yes, in CT liver analysis application cumulative modifications.
d	Have any **"Clarification of Requirements – No Change to Functionality**" been made (changes made to clarify software requirements after a product has received premarket clearance)?	☒ Yes ☐ No	Rationale: Possible clarification or revised phrasing of an existing requirement or creation of a new requirement altogether, without changing or adding functionality were done as part of AVW15 (ISP13). Changes made to clarify the requirements as discussed here likely do not require submission of a new 510(k).
e	Have any **"Cosmetic Changes – No Change to Functionality"** been made (changes made to the appearance of the device that do not impact the clinical use of the device)?	☒ Yes ☐ No	Rationale: Please refer to section 3.3.2

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#	Question	Yes/No	Decision / Rationale
f	Have any **"Reengineering"** and **"refactoring"** changes been made (two common software maintenance techniques. "Reengineering" is defined as the examination and alteration of software to reconstitute it in a new form, and includes the subsequent implementation of the new form)?	☒ Yes ☒ No	Rationale: Please refer to section 3.3.2. Yes to refactoring, No to re-engineering.

9.1.5. Conclusion

According to the flowcharts and tables presented above in section *9.1*:

- It was concluded that only for changes done in CT liver analysis application, CT ASPECTS and CT Calcium Automated analysis which are new applications in AVW15 , 510K is required (currently under FDA review: CT Liver K222233, CT ASPECTS K K213784).

Recommended Regulatory Action: ☒ Submit a 510(k)

- It was concluded that for the list of main modified applications/features described in sections 3.2.6-3.2.15, Internal documentation is required.

Recommended Regulatory Action: ☒ Documentation (Letter to File)

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